

SPIDER

447 series
218 patients
4 hospitals

expert
masks

vendor,
field

R1, R2 intensity

raw signal in masks;
ratios, z scores

R3 mask geometry

volumes, spacing,
counts, voxel size

R4 graph-relational

Laplacian spectrum
+ scale-free ratios



V vertebra
D disc
C canal

Parallel analyses (same protocol)

Scanner predictability

random forest,
grouped CV,
bootstrap,
permutation

Acquisition controls

voxel size,
coverage, sequence,
fixed field,
case mix, ComBat

Downstream utility

degeneration,
narrowing;
within and
across vendors

Series network

k-NN graph (R3),
assortativity,
communities,
patient null